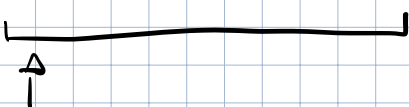


(Static) Range Minimum Queries

Given $A[0, n-1]$

$$RMQ(i, j) = \arg \min (A[i \dots j])$$

A	2	2	0	5	6	7	8	10
pos	0	1	2	3	4	5	6	7

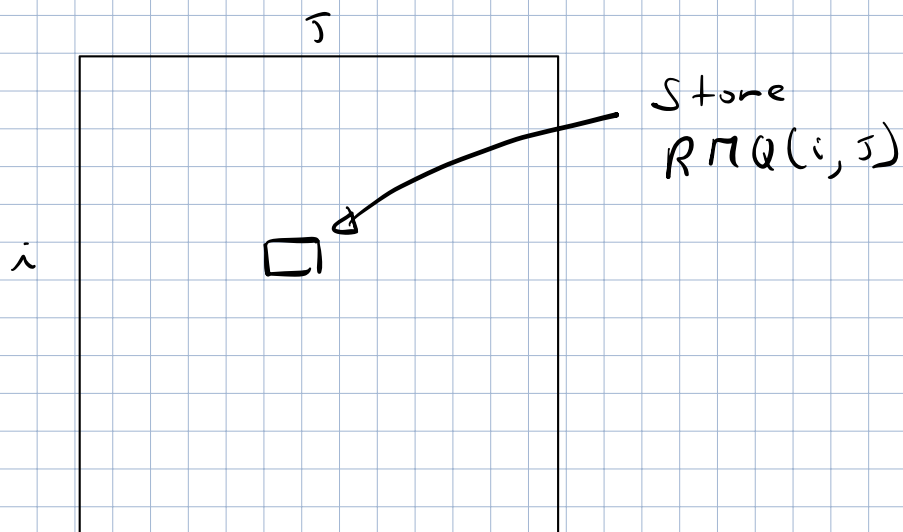


TARGET : $O(1)$ time/query (vs $\Theta(\log n)$ ST)
 $\Theta(n)$ space

Solution #1 $O(1)$ time / $\Theta(n^2)$ space

Store a table with all the possible answers

Table T



Solution #2 Sparse array

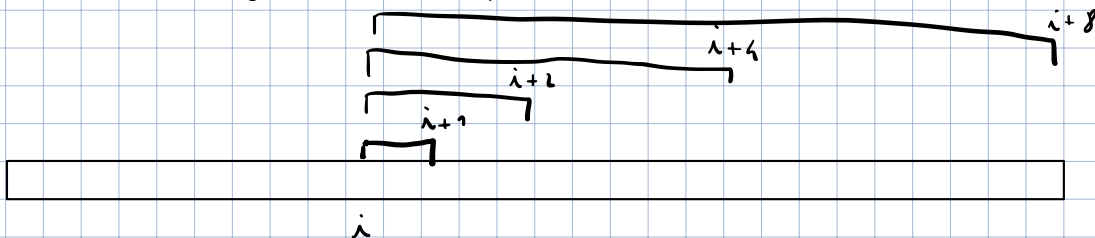
$O(1)$ time / $\Theta(\log n)$ space

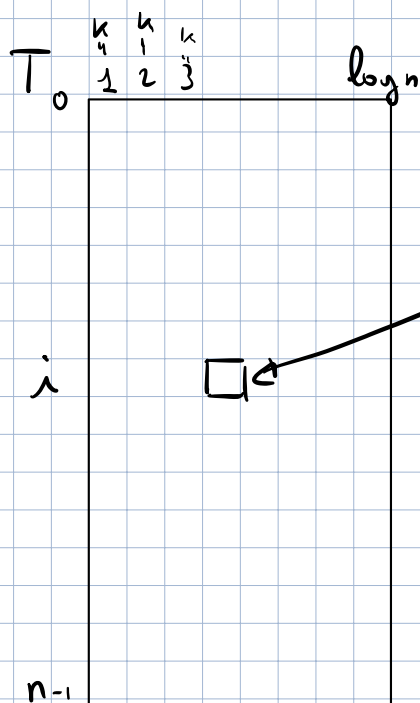
Exploit: $\min(i, j)$ can be solved by computing the min of two overlapping ranges



$$\min(i, j) = \min \begin{cases} \min(i, j') \\ \min(i', j) \end{cases}$$

For any i store the answer for the range $[i, i + 2^k)$ for any k

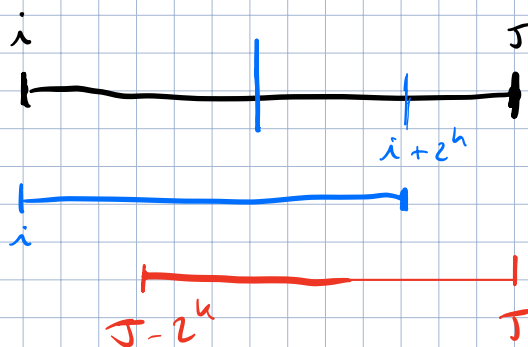




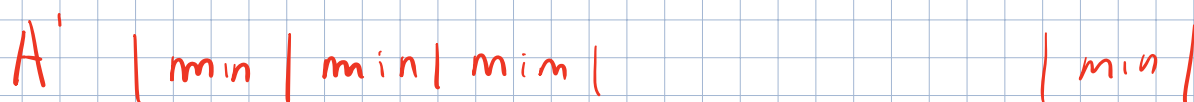
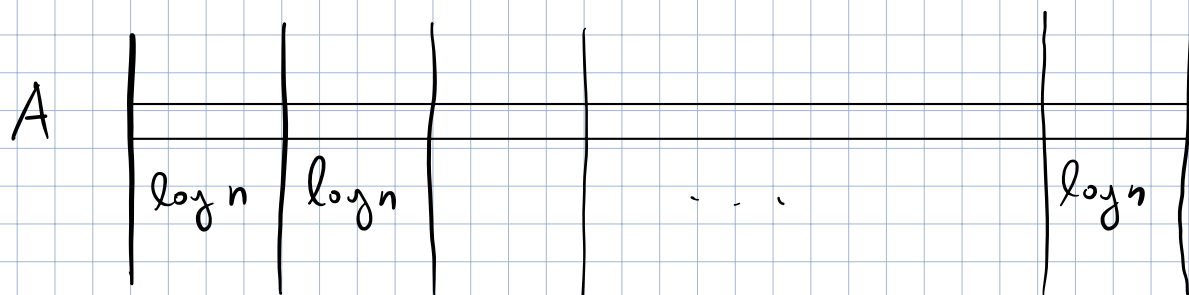
Answer for $[i, i + 2^k]$

$$\text{Space} = \Theta(n \log n)$$

$\text{RMA}(i, j)$



Solution # 3 $\Theta(\log n)$ time / $\Theta(n)$ space

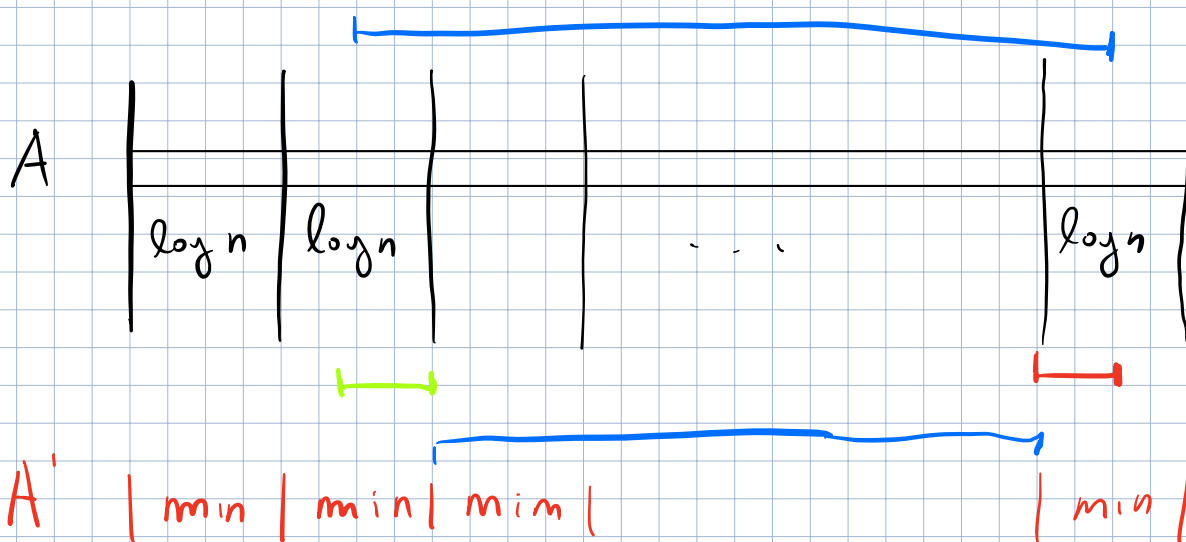


$$|A'| = \frac{n}{\log n}$$

use previous solution

$\Theta(n)$ space

$O(n)$ time



$$RMQ(i, j) = \min \begin{cases} \min & \text{green bracket} \\ \min & \text{blue bracket} \\ \min & \text{red bracket} \end{cases}$$

$\Theta(\log n)$ time

$O(1)$ Time

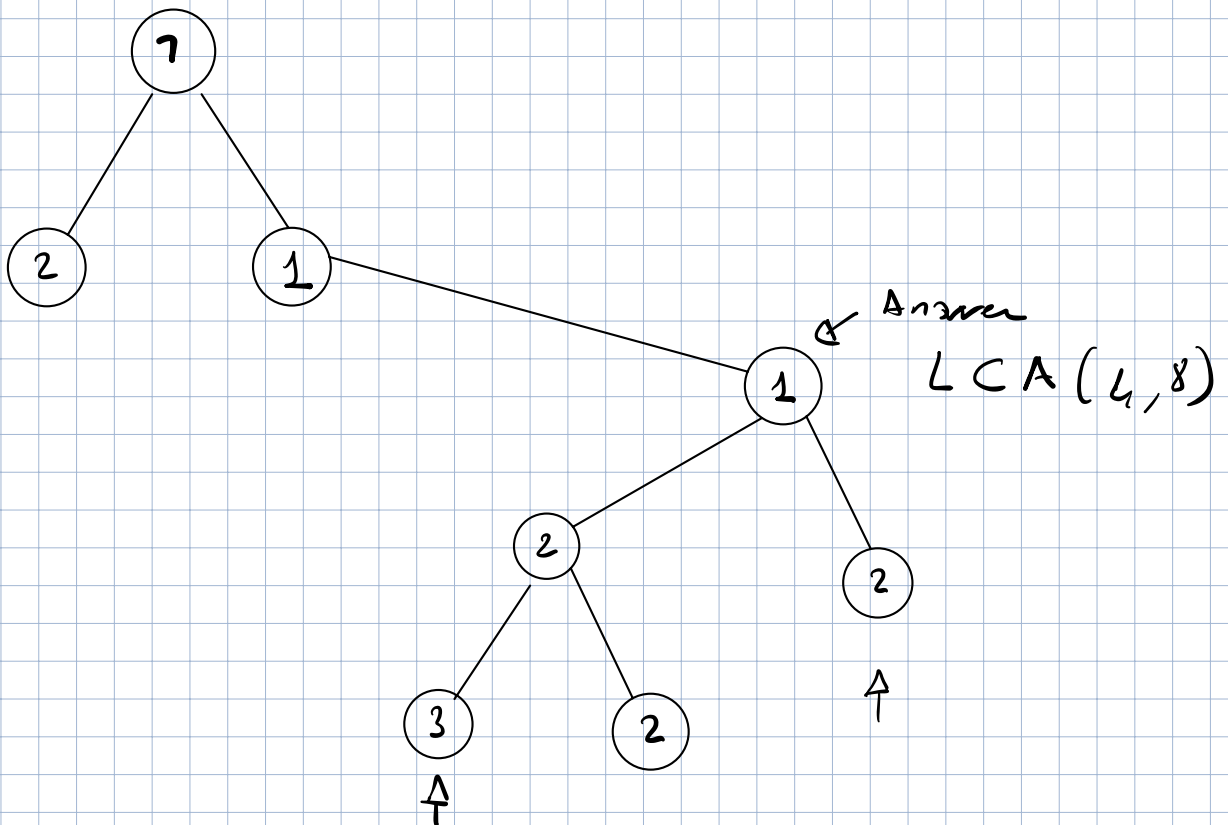
$\Theta(\log n)$ time

Solution #4 $O(1)$ time / $\Theta(n)$ space

Just the intuition

using previous solution, it's enough to solve RMQ on a block of size $\approx \log n$ elements

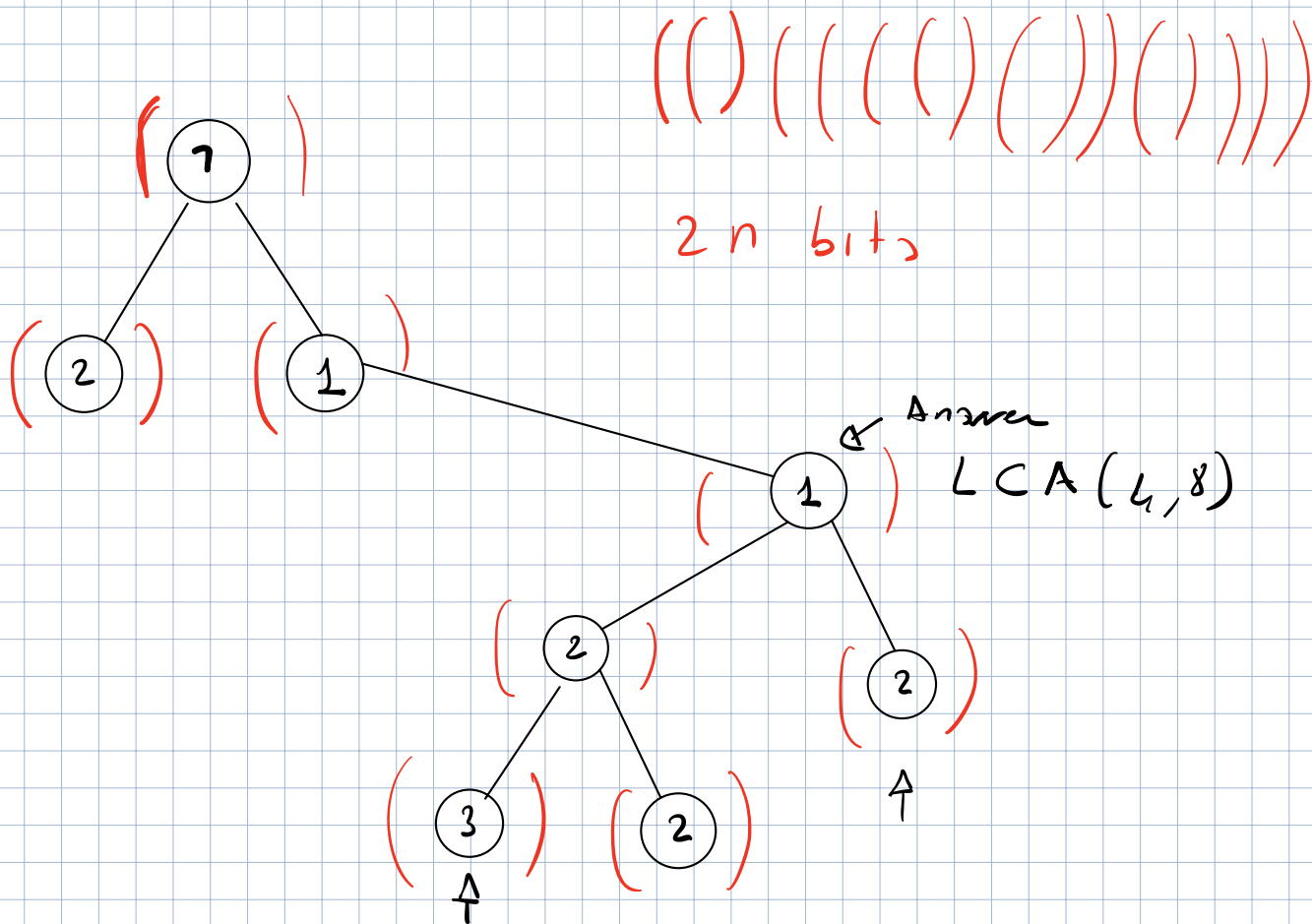
Cartesian tree



A	2	1	1	3	2	2	1	2
pos	1	2	3	4	5	6	7	8

$RMQ(4, 8) =$

, what's this element?



A	2	1	1	3	2	2	1	2
pos	1	2	3	4	5	6	7	8

$$4 \cdot \frac{1}{8} \log n \text{ bits}$$

$$\frac{1}{2} \log n$$

possible tree representation

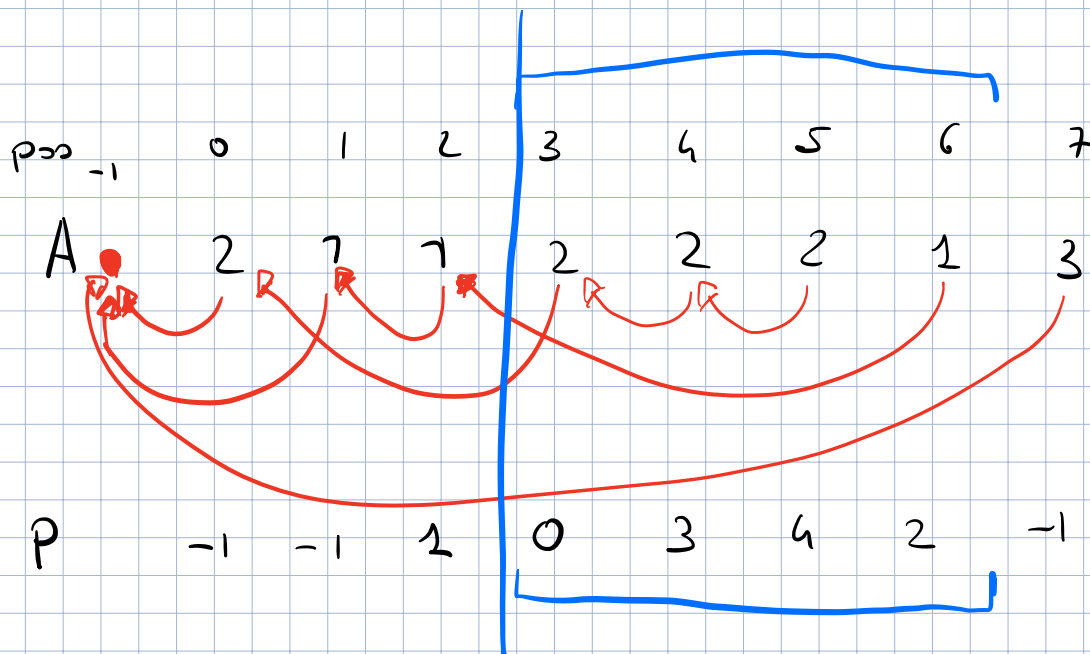
$$= \sqrt{n}$$

- $O(1)$ + 1 me

Solution

space: $\Theta(n)$

query time: $\Theta(k)$



3

